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Data 612 Recommender System – Summer 2025 Project 2

Clarifying Sentiment Analysis & Model Similarity

First, thank you for your feedback on Project 2. I wanted to take a moment to respond to your questions and explain a bit more about how sentiment analysis fits into my recommender and why both models ended up giving similar results, even though they work differently.

1. How Sentiment Analysis Helped the Recommender

This part was essential to me because we're talking about skincare, especially for melanin-rich users who aren't always prioritized in product testing or marketing. Just going by star ratings can be misleading. For example:

- A product might have 4.5 stars, but if reviewers are mentioning burns, rashes, or irritation, that's a big red flag, especially for people with sensitive or underrepresented skin tones.

So, I used sentiment analysis and keyword filtering to catch those kinds of issues:

- The rule-based model filtered out products with negative sentiment or specific harm-related words.
- This gave me a safer recommendation list that wasn't just about popularity, but also about actual user experiences and safety.

2. Why Both Recommenders Gave Similar Results

Even though I used two different recommender types, one collaborative filtering, one rule-based, they both used the same cleaned and filtered dataset. Here's why the results overlapped:

- I narrowed the data down to products rated highly by users with deeper skin tones, so there was already a lot of overlap in what was being selected.
- The models got to the same place for different reasons:
- Collaborative Filtering (CF): Picked similar-user favorites.
- Rule-Based: Ensured products were safe by rules and sentiment.

I saw this as a good sign, as it meant those top products weren't just popular or safe by one metric, but by both. This reinforced the strength and reliability of the recommendations.